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NATIONAL ADVISORY COMMITTEE FOR AERONAUTICS

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SUMMARY OF MEASUREMENTS IN LANGLEY FULL-SCALE TUNNEL OF MAXIMUM LIFT COEFFICIENTS AND STALLING CHARACTERISTICS OF AIRPLANES

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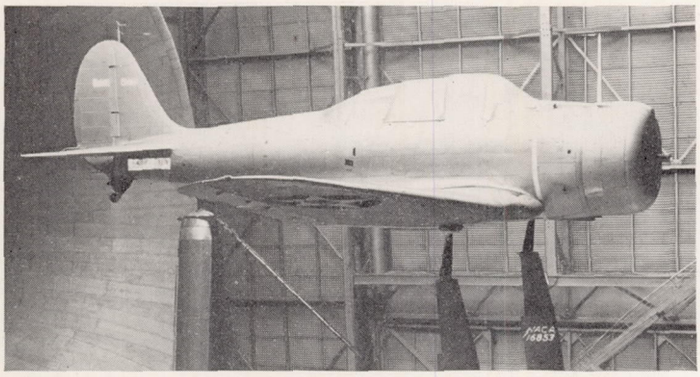
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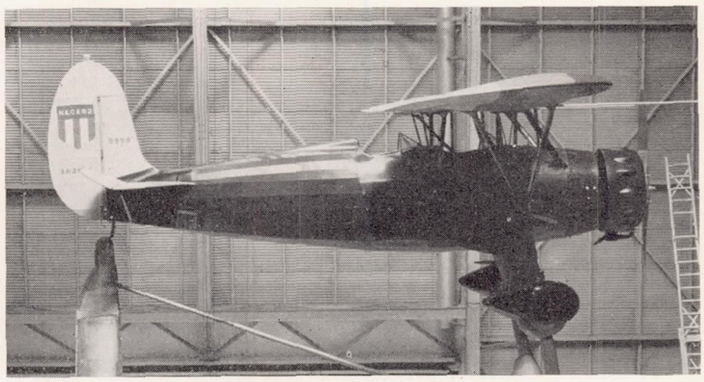
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(q) Airplane 17 with revised canopy.



(r) Airplane 18 in service condition.

FIGURE 2.—Concluded.

TABLE I.—GEOMETRIC CHARACTERISTICS OF AIRPLANES TESTED IN LANGLEY FULL-SCALE TUNNEL

Airplane	Wing area (sq ft)	Aspect ratio	Taper ratio, $\gamma = \frac{\text{Root chord}}{\text{Tip chord}}$	Wing section [a]		Angle of incidence (deg)		Wing flaps			
				Root	Tip	Root	Tip	Type	Average flap chord / Average wing chord	Flap span / Wing span	Maximum flap deflection, $\delta_{f_{max}}$ (deg)
1	233.2	5.89	2.17	NACA–NAA compromise low drag	NACA–NAA compromise low drag	1.00	−1.25	Slotted	24.8	50.7	50.0
2	248.0	5.93	2.00	NACA 66(2x15)−116, a=0.6	NACA 66(2x15)−216, a=0.6	1.30	−.45	Plain	16.6	31.3	45.0
3	100.0	7.56	3.00	NACA 65(216)−017	NACA 67,1−(1.3)15, a=0.7	2.00	.50	do	22.1	47.0	60.0
4	284.0	5.30	1.81	NACA 66(215)−114	NACA 66(215)−213, a=0.6	1.71	−.80	Slotted	25.0	51.5	45.0
5	334.0	5.50	2.00	NACA 23016	NACA 23009	3.00	3.00	do	25.6	64.1	48.0
6	314.0	5.30	1.47	NACA 23018	NACA 23009	2.00	2.00	do	22.6	56.4	50.0
7	422.0	5.90	2.32	NACA 23017	NACA 23009	1.50	−.50	Split	23.7	61.1	60.0
8	223.7	5.90	[b]	NACA 0015	NACA 0009	1.00	1.00	do	32.0	53.3	60.0
9	250.0	5.19	4.00	NACA 66,2−018	NACA 66,2−018	0	0				
10	203.4	8.10	3.38	CW 6500−0015	CW 6500−0015	2.00	−1.50	Split	23.6	36.0	45.0
11	208.9	6.00	1.40	NACA 23018	NACA 23009	0	0	Slotted	26.6	100.0	50.6
12	233.2	5.00	1.48	NACA 23015	NACA 23009	0	0	Split	25.0	51.5	60.0
13	172.0	8.10	4.00	NACA 0018	NACA 0010	4.60	4.60	do	20.0	57.7	60.8
14	148.0	11.90	3.35	NACA 23024	NACA 23009	5.50	5.50	Slotted	[c] 28.0	[c] 53.4	55.0
15	188.0	7.70	2.50	NACA 0018	NACA 0009	2.00	2.00	Split	23.1	57.8	50.0
16	170.0	5.70	1.76	NACA 23015	NACA 23009	1.00	1.00	Slotted	25.7	55.2	40.0
17	318.6	5.40	1.96	NACA 2415	NACA 2409	2.50	2.50	Split	23.7	65.6	45.0
18	180.0	6.00	1.00	NACA $2R_112$	NACA $2R_112$	0	0				

[a] The designations of the NACA low-drag airfoils have been changed from the form furnished by the manufacturer to the form described in reference 15.
[b] Elliptical chord distribution.
[c] Only inboard and center flaps deflected.

Most of the measurements were made at tunnel airspeeds of approximately 60 miles per hour; a few tests were made at slightly lower airspeeds. In order to indicate the effect of variation in Reynolds number, measurements were made for some of the airplanes over an approximate range of tunnel velocity from 20 to 100 miles per hour.

Force readings were taken for one of the airplanes (airplane 18) at regular intervals while the angle of attack was being changed at a constant rate in order to obtain a comparison with flight measurements of maximum lift coefficient. The rate of change of angle of attack per second for these tests was varied between 0.025° and 0.200°.

The usual wind-tunnel jet-boundary and blocking corrections have been applied to all the data.

RESULTS AND DISCUSSION

The results of measurements of maximum lift coefficients and stalling characteristics of 18 airplanes tested in the Langley full-scale tunnel are summarized in the following sections. In most cases the results are given for the airplanes with landing flaps retracted and with landing flaps fully extended. The data are grouped in the first five sections to show the characteristic effects on maximum lift and stall of wing geometry, fuselages and nacelles, propeller slipstream, surface roughness and leakage, and wing leading edge appendages. In the final sections, comparisons are made of the increments of lift coefficient due to split and slotted flaps and of wind-tunnel and flight measurements of maximum lift coefficients of airplanes.

WING GEOMETRY

Conventional plan forms.—Stall progressions for airplanes with untwisted wings of different taper ratios (airplanes 13, 12, and 8) are presented in figure 3 for landing flaps retracted and fully deflected. Although these data are given for complete airplanes with fuselages and nacelles but with propellers removed, the results show trends generally characteristic of the effects of wing taper ratio on the progression of the stall.

With the landing flaps retracted (fig. 3(a)), local areas of separation appeared on airplane 13 (wing taper ratio, 4:1) at the wing trailing edge near the fuselage and behind oil-cooler outlets located just outboard of each nacelle for relatively low angles of attack; the main stall, however, started at the wing tips and progressed inboard with increasing angle of attack. Theoretical studies (references 2 to 4) show that, for plain untwisted wings of high taper ratio, the section

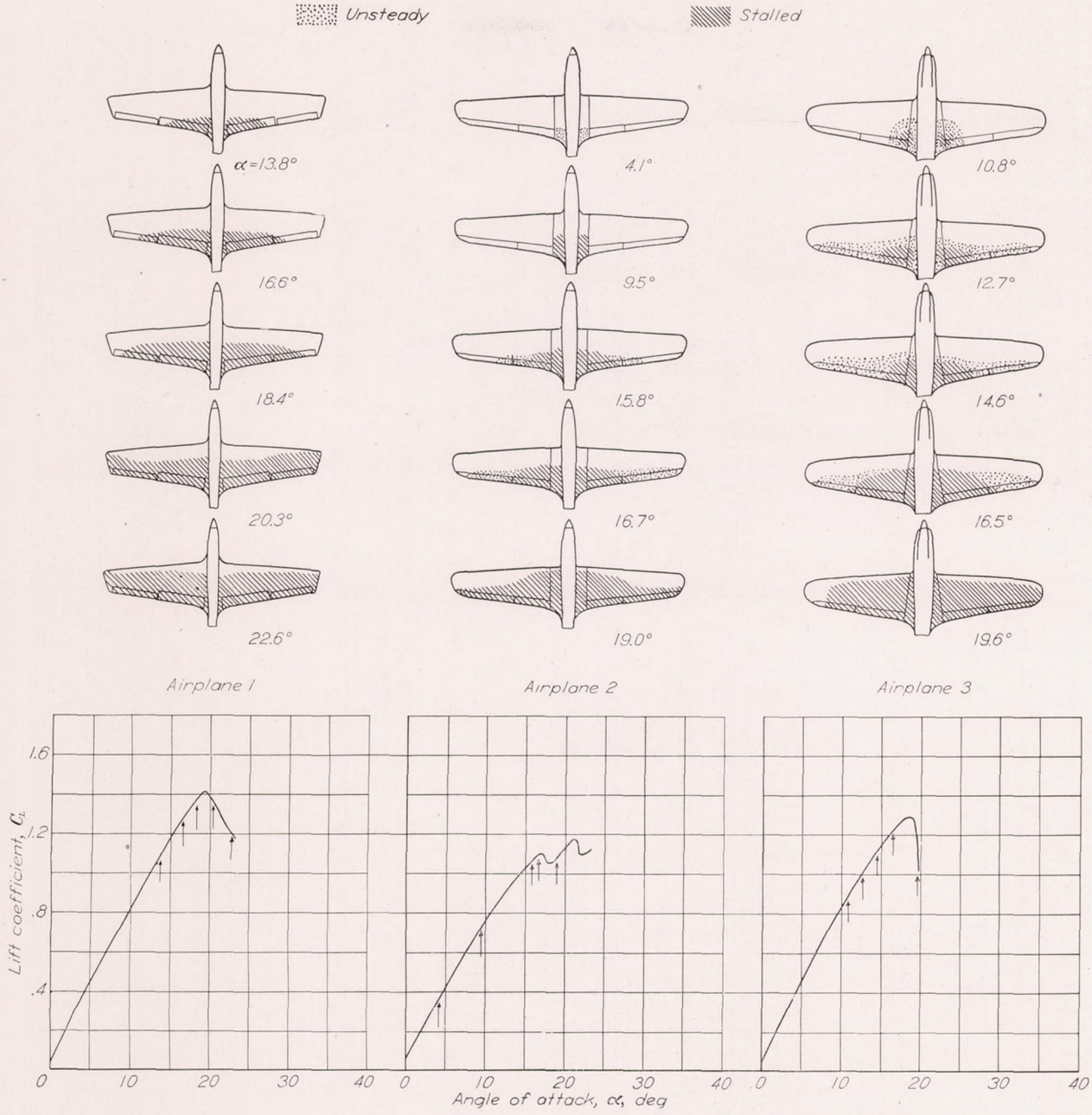


FIGURE 4.—Stall progressions for three present-day airplanes having low-drag wings. Complete airplanes less propellers; $\delta_f=0°$; approximate test velocity, 60 miles per hour.

The exact nature of the effects of flap deflection on the stalling characteristics of airplanes is not well defined. Flight observations of a large number of airplanes tested in the United States and in England (references 5 and 6) have indicated that flap deflection either improved or aggravated the stall in about an equal number of cases. Flap deflection generally tends to aggravate the stall by increasing the upwash over the outer unflapped parts of the wing and by cleaning up the area of separation at the root. On the other hand, the handling characteristics of an airplane in flight near the stall may be improved by flap deflection if the flap wake envelops the tail at angles of attack near the stall and thus produces a stall warning either by tail buffeting or by a rapid change in trim due to the loss in tail effectiveness.

Stall progressions for three typical present-day pursuit airplanes having twisted wings of low-drag airfoil sections (airplanes 1, 2, and 3) are shown in figure 4. The taper ratio and washout of the wings of these three airplanes are nearly the same. (See table I for wing details.) The stalls are strikingly similar; separation begins, in each case, at the wing-fuselage juncture and progresses outboard along the rearward portion of the wing with increasing angle of attack. The stalling characteristics of these airplanes, as interpreted from the tuft observations, are probably good. Although airplane 3 shows a rapid loss in lift after the stall, no serious trouble should be encountered by the pilot inasmuch as the root-section stall should provide adequate warning of the approach of $C_{L_{max}}$.